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AoA Detection Using a Single Rydberg Atomic Receiver: Leveraging Inner-Vapor Interference

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March 26, 2026



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Abstract—Rydberg atomic receivers have been envisaged as a revolutionary technology for future wireless communications and sensing. In order to detect the angle of arrival (AoA), researchers have typically constructed arrays comprising *multiple* Rydberg atomic receivers. This paper presents a novel finding: The AoA of an incident signal can be accurately recovered even with a *single* Rydberg atomic receiver, by harnessing the phenomenon of inner-vapor interference. Firstly, we apply the micro-element method to the atomic vapor and derive a closed-form expression for the laser transmission in the presence of interference between the incident and local oscillator (LO) radio frequency (RF) signals. Secondly, we propose a robust method to estimate the AoA based on the particle swarm optimization (PSO) algorithm. Simulation results substantiate the effectiveness of our proposed scheme with practical parameter settings, verifying the applicability of AoA detection using a single Rydberg atomic receiver.

Index Terms—Rydberg atomic receiver, Rydberg atom, angle of arrival (AoA), quantum sensing, interference.

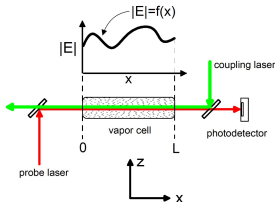
A. P. Rotunno et al., “Investigating electromagnetically induced transparency spectral lineshape distortion due to non-uniform fields in Rydberg-atom electrometry,” J. Appl. Phys., Aug. 2023.

► Contribution:

$$T = \frac{P_{\text{out}}}{P_{\text{in}}} = \exp \left(-\frac{2\pi L}{\lambda_p N_S} \sum_{i=1}^{N_S} \text{Im}(\chi(x_i)) \right),$$

T is the transmittance, as the ratio of output power P_{out} to input power P_{in} , L is the length of the medium, λ_p is the wavelength, N_S is the number of sampling points, and $\chi(x_i)$ is the complex susceptibility at position x_i , with Im denoting the imaginary part.

► Cited: 33



Y. Cao, T. Lv, and W. Ni, “Two-timescale optimization for intelligent reflecting surface-assisted MIMO transmission in fast-changing channels,” *IEEE Trans. Wireless Commun.*, vol. 21, no. 12, pp. 10 424–10 437, Dec. 2022.

► Contribution:

We propose a new PSO framework to solve Problem (P3). The PSO, a widely-used meta-heuristic bionic optimization algorithm [15], can find the optimal solution with little prior information. It enjoys the benefits of fewer parameters, simple calculation implementation, and fast convergence [44]. In the

$$\mathbf{v}_p^{(i)} = w\mathbf{v}_p^{(i-1)} + c_1\varepsilon_1(\dot{\boldsymbol{\theta}}_p^{(i-1)} - \boldsymbol{\theta}_p^{(i-1)}) + c_2\varepsilon_2(\ddot{\boldsymbol{\theta}}_p^{(i-1)} - \boldsymbol{\theta}_p^{(i-1)}), \quad (19)$$

$$\boldsymbol{\theta}_p^{(i)} = \boldsymbol{\theta}_p^{(i-1)} + \mathbf{v}_p^{(i)}, \quad (20)$$

► Cited: 32

- ▶ **Precise Dynamics of Rydberg Atomic Receiver:** Modeling the Rydberg atom as a sophisticated four-energy-level system, derive a closed-form expression for the susceptibility of atomic vapor by solving the initial equation under practical constraints.
- ▶ **Quantifying Inner-Vapor Interference:** Using this approach, we unveil that the output probe laser power is correlated with the AoA of the incident signal in the presence of LO signals.
- ▶ **Efficient Method for AoA Detection:** Proposing an efficient particle swarm optimization (PSO)-based methodology to address the significant non-convexity of the residual function.
- ▶ **Determination of Parameter Settings:** Through meticulous theoretical analysis and comprehensive simulation results, the selection of some crucial parameters is determined.
- ▶ **Successful AoA Detection:** The root mean square error (RMSE) of AoA detection can be less than 1 degree when the number of measurements is larger than 5 and the signal-to-noise ratio (SNR) is larger than 60 dB.

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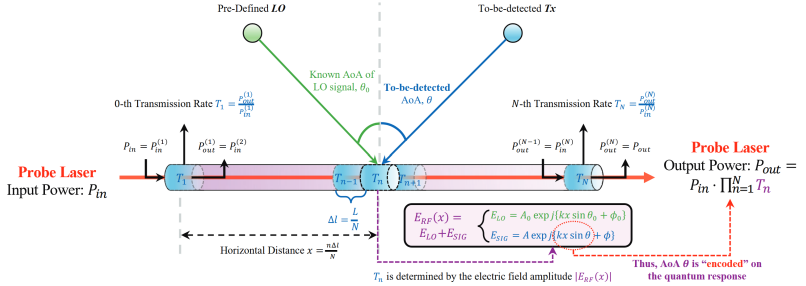
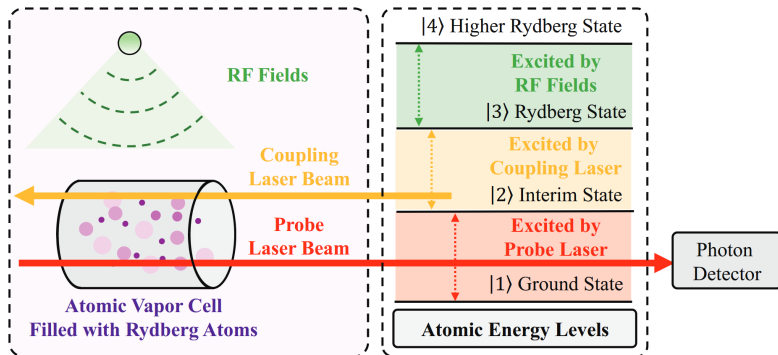


Fig. 3: Illustration of the micro-element method. The atomic vapor is uniformly divided into N micro-blocks ($N \gg 1$). When the length of each micro-block approaches 0, the product of the transmission coefficients in (1) can be transformed into the integral in (17) according to Riemann's theorem.

- ▶ when the length of the atomic vapor cell is comparable to the wavelength, the spatially non-uniform electric field magnitude across the atomic vapor cell can lead to the spatially non-uniform probe laser transmission.
- ▶ divide the atomic vapor cell of length L into N microblocks uniformly,

$$P_{out} = P_{in} \prod_{n=1}^N T_n. \quad (1)$$



- Introduce energy level and its physics
- detected physical quantity- $P \rightarrow T \rightarrow \chi \rightarrow \rho_{12} \rightarrow E$

Introduce the Lindblad Liouville-von Neumann equation, $\rho \in \mathbb{C}^{4 \times 4}$

$$\frac{d\rho}{dt} = -\frac{j}{h}(\mathbf{H}\rho - \rho\mathbf{H}) + \mathbf{L}, \quad \rho = \begin{pmatrix} \rho_{11} & \rho_{12} & \rho_{13} & \rho_{14} \\ \rho_{21} & \rho_{22} & \rho_{23} & \rho_{24} \\ \rho_{31} & \rho_{32} & \rho_{33} & \rho_{34} \\ \rho_{41} & \rho_{42} & \rho_{43} & \rho_{44} \end{pmatrix}, \quad (2)$$

$$\mathbf{H} = h \begin{bmatrix} 0 & \frac{\Omega_p}{2} & 0 & 0 \\ \frac{\Omega_p}{2} & -\Delta_p & \frac{\Omega_c}{2} & 0 \\ 0 & \frac{\Omega_c}{2} & -(\Delta_p + \Delta_c) & \frac{\Omega_{RF}}{2} \\ 0 & 0 & \frac{\Omega_{RF}}{2} & -(\Delta_p + \Delta_c + \Delta_{RF}) \end{bmatrix}, \quad E = hf = \hbar\omega \quad (3)$$

$$\mathbf{L} = \begin{bmatrix} \Gamma_2\rho_{22} & -\gamma_{12}\rho_{12} & -\gamma_{13}\rho_{13} & -\gamma_{14}\rho_{14} \\ -\gamma_{21}\rho_{21} & -\Gamma_2\rho_{22} + \Gamma_3\rho_{33} & -\gamma_{23}\rho_{23} & -\gamma_{24}\rho_{24} \\ -\gamma_{31}\rho_{31} & -\gamma_{32}\rho_{32} & -\Gamma_3\rho_{33} + \Gamma_4\rho_{44} & -\gamma_{34}\rho_{34} \\ -\gamma_{41}\rho_{41} & -\gamma_{42}\rho_{42} & -\gamma_{43}\rho_{43} & -\Gamma_4\rho_{44} \end{bmatrix}. \quad (4)$$

$$\begin{cases} \dot{\rho}_{21} = 0 \\ \dot{\rho}_{31} = 0 \\ \dot{\rho}_{41} = 0 \end{cases} \Rightarrow \begin{cases} -j \left(\frac{\Omega_p \rho_{11}}{2} - \Delta_p \rho_{21} + \frac{\Omega_c \rho_{31}}{2} - \frac{\Omega_p \rho_{22}}{2} \right) = \gamma_{21} \rho_{21}, \\ -j \left[\frac{\Omega_c \rho_{21}}{2} - (\Delta_p + \Delta_c) \rho_{31} + \frac{\Omega_{RF} \rho_{41}}{2} - \frac{\Omega_p \rho_{32}}{2} \right] = \gamma_{31} \rho_{31}, \\ -j \left[\frac{\Omega_{RF} \rho_{31}}{2} - (\Delta_p + \Delta_c + \Delta_{RF}) \rho_{41} - \frac{\Omega_p \rho_{42}}{2} \right] = \gamma_{41} \rho_{41}. \end{cases} \quad (5)$$

- introduce equation's symbols and their physics meanings.

Theorem 1: Weak Probe Field Condition

- ▶ When the probe laser is relatively weak, the probability that the atom stays in $|1\rangle$ can be approximated as 1, which indicates $\rho_{11} \approx 1$ and $\rho_{22} \approx \rho_{33} \approx \rho_{44} \approx 0$.
- ▶ Moreover, when the probe laser is relatively weak, the influence of ρ_{32} and ρ_{42} on ρ_{21} can be disregarded as illustrated in Fig. 21.

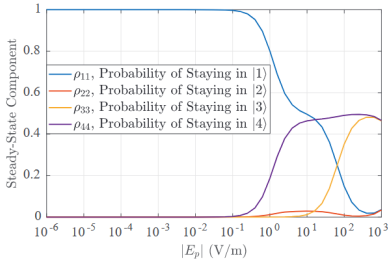


Fig. 20: ρ_{11} , ρ_{22} , ρ_{33} , and ρ_{44} versus the electric field magnitude of the probe beam $|E_p|$. Other parameters are $\Delta_p = \Delta_c = \Delta_{RF} = 0$, $\Omega_c = 40$ MHz, and $\Omega_{RF} = 12$ MHz [18]. As can be observed, when the probe laser is weak (e.g. $|E_p| < 0.1$ V/m in the figure), such as the 360 μ W probe laser power adopted in [48], we have $\rho_{11} \approx 1$ and $\rho_{22} \approx \rho_{33} \approx \rho_{44} \approx 0$.

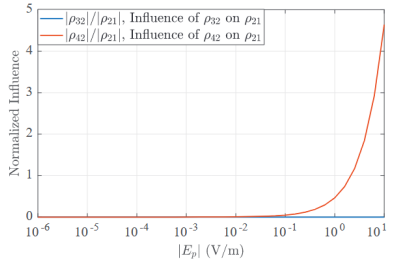


Fig. 21: ρ_{32} and ρ_{42} versus the electric field magnitude of the probe beam $|E_p|$. Other parameters are $\Delta_p = \Delta_c = \Delta_{RF} = 0$, $\Omega_c = 40$ MHz, and $\Omega_{RF} = 12$ MHz [18]. As can be observed, when the probe laser is weak (e.g. $|E_p| < 0.1$ V/m in the figure), such as the 360 μ W probe laser power adopted in [48], the influence of ρ_{32} and ρ_{42} on ρ_{21} can be disregarded.

$$\begin{cases} -j \left(\frac{\Omega_p}{2} - \Delta_p \rho_{21} + \frac{\Omega_c \rho_{31}}{2} \right) = \gamma_{21} \rho_{21}, \\ -j \left[\frac{\Omega_c \rho_{21}}{2} - (\Delta_p + \Delta_c) \rho_{31} + \frac{\Omega_{RF} \rho_{41}}{2} \right] = \gamma_{31} \rho_{31}, \\ -j \left[\frac{\Omega_{RF} \rho_{31}}{2} - (\Delta_p + \Delta_c + \Delta_{RF}) \rho_{41} \right] = \gamma_{41} \rho_{41}. \end{cases} \quad (6)$$

$$\rho_{41} = \frac{-j \frac{\Omega_{RF}}{2}}{\gamma_{41} - j(\Delta_p + \Delta_c + \Delta_{RF})} \rho_{31}, \quad (7)$$

$$\rho_{31} = \frac{-j \frac{\Omega_c}{2}}{\gamma_{31} - j(\Delta_p + \Delta_c) + \frac{\Omega_{RF}^2}{4 [\gamma_{41} - j(\Delta_p + \Delta_c + \Delta_{RF})]}} \rho_{21}, \quad (8)$$

$$\rho_{21} = \frac{-j \frac{\Omega_p}{2}}{\gamma_{21} - j\Delta_p + \frac{\Omega_c^2}{4} \frac{1}{\gamma_{31} - j(\Delta_p + \Delta_c) + \frac{\Omega_{RF}^2}{4} \frac{1}{\gamma_{41} - j(\Delta_p + \Delta_c + \Delta_{RF})}}} \quad (9)$$

Theorem 2: Tiny Decay Rate Condition

- $\gamma_{31} \approx \gamma_{41} \approx 0$, which is determined by the long lifetime and stability of the Rydberg state.
- In addition, Δ_p could be set to zero.

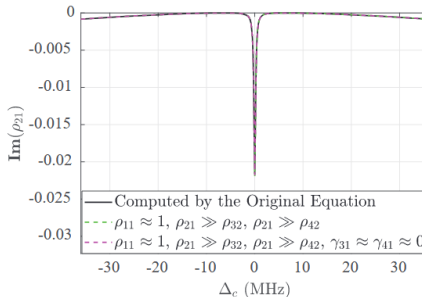


Fig. 22: The imaginary part of ρ_{21} computed by the original equation (25), and that after approximation $\rho_{11} \approx 1$, $\rho_{21} \approx \rho_{32}$, $\rho_{21} \approx \rho_{42}$, and $\gamma_{31} \approx \gamma_{41} \approx 0$. The normalized mean square error after applying this approximation is less than 40 dB, suggesting the correctness of our proposed approximation.

$$\rho_{21} = \frac{-j \frac{\Omega_p}{2}}{\gamma_{21} + \frac{\Omega_c^2/4}{-j\Delta_c + \frac{\Omega_{RF}^2/4}{-j(\Delta_c + \Delta_{RF})}}} \quad (10)$$

In conclusion, limitations on above derivation include:

- ▶ No collision terms or dephasing terms: Rydberg-atom collisions, Penning ionization, and ion electric fields can, in principle, cause dephasing, such effects can, for instance, be alleviated by reducing the beam intensities, lowering the vapor pressure, or limiting the atom-field interaction time;
- ▶ Weak probe approximation, $\rho_{11} \approx 1$ and $\rho_{22} \approx \rho_{33} \approx \rho_{44} \approx \rho_{42} \approx \rho_{32} \approx 0$;
- ▶ Long life of Rydberg state causes small transition decay rates, $\gamma_{31} \approx \gamma_{41} \approx 0$.

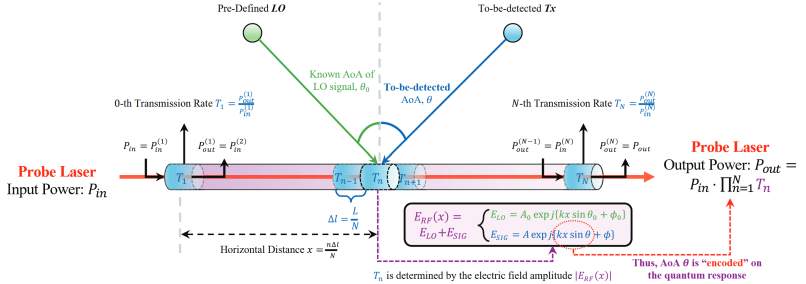


Fig. 3: Illustration of the micro-element method. The atomic vapor is uniformly divided into N micro-blocks ($N \gg 1$). When the length of each micro-block approaches 0, the product of the transmission coefficients in (1) can be transformed into the integral in (17) according to Riemann's theorem.

$$\Omega_{RF}(x) = \frac{|E_{RF}(x)|}{h} \mu_{RF}, \quad (11)$$

$$E_{RF}(x) = A_0 e^{j(kx \sin \theta_0 + \phi_0)} + A e^{j(kx \sin \theta + \phi)}, \quad (12)$$

and μ_{RF} denotes the dipole moment of the RF transition.

$$\Omega_{RF}(x) = \frac{|A_0 e^{j(kx \sin \theta_0 + \phi_0)} + A e^{j(kx \sin \theta + \phi)}|}{h} \mu_{RF} \quad (13)$$

$$= \frac{|e^{j\phi_0} [A_0 e^{jkx \sin \theta_0} + A e^{jk(x \sin \theta + \Delta \phi)}]|}{h} \mu_{RF} \quad (14)$$

$$= \frac{\sqrt{A_0^2 + A^2 + 2A_0 A \cos [(k \sin \theta_0 - k \sin \theta)x - \Delta \phi]}}{h} \mu_{RF} \quad (15)$$

$$\rho_{21}(x) = \frac{-j \frac{\Omega_P}{2}}{\gamma_{21} + \frac{\Omega_c^2/4}{-j\Delta_c + \frac{A_0^2 + A^2 + 2A_0 A \cos [(k \sin \theta_0 - k \sin \theta)x - \Delta \phi]}{4h^2 \mu_{RF}^{-2} [-j(\Delta_c + \Delta_{RF})]}}} \quad (16)$$

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In uniform field,

$$P_{out} = P_{in} \exp \left(-\frac{2\pi L \text{Im} [\chi]}{\lambda_p} \right) = P_{in} \exp (-\alpha L), \quad (17)$$

$\alpha = 2\pi \Im[\chi]/\lambda_{pr} = k_{pr} \Im[\chi]$ is Beer's absorption coefficient for the probe laser, where $k_{pr} = 2\pi/\lambda_{pr}$ denotes the wavenumber of the probe laser.

The susceptibility for the probe laser is related to the density matrix component (ρ_{21}) by the following expression,

$$\chi(x) = -\frac{4\pi N_a \mu_p^2}{\epsilon_0 h \Omega_p} \rho_{21}(x), \quad (18)$$

- ▶ where N_a denotes the density of atoms in the atomic vapor.
- ▶ $\mu_p = 2qa_0$ denotes the dipole moment of the ground-state transition for the probe laser, where $q = 1.6 \times 10^{-19}$ C is the charge of an electron and $a_0 = 5.292 \times 10^{-11}$ m is the Bohr radius .
- ▶ $\epsilon_0 = 8.84 \times 10^{-12}$ F/m denotes the permittivity in vacuum.
- ▶ $\rho_{21}(x)$ denotes the density matrix component at location x under the steady-state condition.

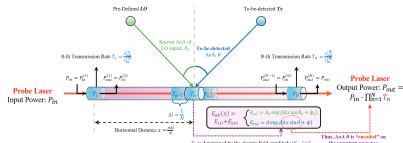


Fig. 3: Illustration of the micro-element method. The atomic vapor is uniformly divided into N micro-blocks ($N \gg 1$). When the length of each micro-block approaches 0, the product of the transmission coefficients in (1) can be transferred into the integral in (17) according to Riemann's theorem.

$$P_{out} = P_{in} \prod_{n=1}^N T_n, \quad (19)$$

$$T_n = T\left(\frac{nL}{N}\right) = \exp\left\{-k_{pr} \frac{L}{N} \text{Im}\left[\chi\left(\frac{nL}{N}\right)\right]\right\}, \quad (20)$$

$$T(A, \theta, \Delta\phi, \Delta_c) = \exp\left\{-\frac{2k_{pr}\pi N_a \mu_p^2 \gamma_{21}}{\epsilon_0 h} I(A, \theta, \Delta\phi, \Delta_c)\right\}, \quad (21)$$

$$I(A, \theta, \Delta\phi, \Delta_c) = \int_0^L \left[\gamma_{21}^2 + \left(\frac{\Omega_c^2/4}{\Delta_c - \frac{\{A_0^2 + A^2 + 2A_0 A \cos[k(\sin\theta_0 - \sin\theta)x - \Delta\phi]\}}{4h^2 \mu_{RF}^{-2}(\Delta_c + \Delta_{RF})}} \right)^2 \right]^{-1} dx \quad (22)$$

- Based on these considerations, we aim to estimate the unknown parameters $\{A, \theta, \Delta\phi\}$ from the pre-specified parameters $\{A_0, L, \theta_0\}$, the known coupling laser frequency detuning Δ_c and the observed probe laser transmission coefficient $T = P_{out}/P_{in}$.
- To be specific, we denote the frequency detunings of the coupling laser as $\{\Delta_{c,1}, \Delta_{c,2}, \dots, \Delta_{c,M}\}$. The probe transmission values measured under these frequency detunings are denoted as $\{T_1, T_2, \dots, T_M\}$. $w_m \sim \mathcal{CN}(0, \sigma_w^2)$ denotes the photon-shot noise, which is an additive white Gaussian noise (AWGN).

$$T_m = T(A, \theta, \Delta\phi, \Delta_{c,m}) + w_m, m \in \{1, 2, \dots, M\}, \quad (23)$$

$$\{\hat{A}, \hat{\theta}, \hat{\Delta\phi}\} = \arg \min_{A, \theta, \Delta\phi} R(A, \theta, \Delta\phi), \quad (24)$$

$$R(A, \theta, \Delta\phi) = \sum_{m=1}^M [T_m - T(A, \theta, \Delta\phi, \Delta_{c,m})]^2. \quad (25)$$

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Algorithm 1 PSO-Based Estimation

Input: Frequency detunings $\Delta_{c,m}$ and the corresponding probe laser transmission values T_m for $m \in \{1, 2, \dots, M\}$, number of iterations N_{iter} , number of particles N_{part} , inertia weight w , cognitive parameter c_1 , social parameter c_2 .

Output: Estimated parameters of the incident signal $\{\hat{A}, \hat{\theta}, \hat{\Delta\phi}\}$.

- 1: **Step 1: Particle Initialization**
- 2: **for** $i = 1$ to N_{part} **do**
- 3: Randomly initialize $\mathbf{p}_{\text{temp}}^{(i)}$ and $\mathbf{v}^{(i)}$.
- 4: Initialize $\mathbf{p}_{\text{pbest}}^{(i)} \leftarrow \mathbf{p}_{\text{temp}}^{(i)}$.
- 5: **end for**
- 6: Initialize $\mathbf{p}_{\text{global}} \leftarrow \arg \min_{\mathbf{p}_{\text{pbest}}^{(i)}} \left\{ R \left(\mathbf{p}_{\text{pbest}}^{(i)} \right) \right\}$.
- 7: **Step 2: Iterative Search**
- 8: **for** $n_{\text{iter}} = 1$ to N_{iter} **do**
- 9: **for** $i = 1$ to N_{part} **do**
- 10: Update the particle velocity $\mathbf{v}^{(i)}$ via (21).
- 11: Update the particle position $\mathbf{p}_{\text{temp}}^{(i)}$ via (22).
- 12: **if** $R \left(\mathbf{p}_{\text{temp}}^{(i)} \right) < R \left(\mathbf{p}_{\text{pbest}}^{(i)} \right)$ **then**
- 13: Update $\mathbf{p}_{\text{pbest}}^{(i)} \leftarrow \mathbf{p}_{\text{temp}}^{(i)}$.
- 14: **end if**
- 15: **end for**
- 16: Update $\mathbf{p}_{\text{global}} \leftarrow \arg \min_{\mathbf{p}_{\text{pbest}}^{(i)}} \left\{ R \left(\mathbf{p}_{\text{pbest}}^{(i)} \right) \right\}$.
- 17: **end for**
- 18: **return** $\{\hat{A}, \hat{\theta}, \hat{\Delta\phi}\} \leftarrow \mathbf{p}_{\text{global}}$.

We utilize N_{swarm} particles to explore the minimum within the constrained search space.

- ▶ $P_{\text{temp}}^{(i)} = (A, \theta, \Delta\phi)$ represents the i -th particle, corresponding to a solution within the feasible solution set of the objective function.
- ▶ $P_{\text{pbest}}^{(i)}$ stores the historical optimal solution of the i -th particle (most likely a local optimum; if a global optimum exists, it will converge).
- ▶ P_{global} stores the historical optimal solution of all particles as a whole (if a global optimum exists, it will converge).

The update direction of a solution (particle) is regulated by three factors: motion inertia (exploring the unknown search space), individual experience (historical optimum of a single particle), and socially shared information (historical optimum of all particles).

$$P_{\text{temp}}^{(i)} \leftarrow P_{\text{temp}}^{(i)} + v^{(i)}. \quad (26)$$

$$v_{\text{new}}^{(i)} \leftarrow w v_{\text{old}}^{(i)} + c_1 \epsilon_1 \left[P_{\text{pbest}}^{(i)} - P_{\text{temp}}^{(i)} \right] + c_2 \epsilon_2 \left[P_{\text{global}} - P_{\text{temp}}^{(i)} \right], \quad (27)$$

where w , c_1 and c_2 denote the inertia weight, cognitive parameter, and social parameter, respectively. ϵ_1 and ϵ_2 are random variables distributed as $U(0, 1)$.

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Parameter	Description	Value / Expression
h	Planck constant	6.63×10^{-34} J/Hz
ϵ_0	Permittivity of vacuum	8.85×10^{-12} F/m
N_a	Density of atoms in atomic vapor	4.13×10^{13} m $^{-3}$
λ_{pr}	Probe laser wavelength	780.24 nm
λ_{cp}	Coupling laser wavelength	479.32 nm [18, Fig. 5]
$\nu_4 - \nu_3$	Intrinsic frequency of $ 3\rangle \leftrightarrow 4\rangle$	2.03 GHz [18, Fig. 5]
μ_p	Dipole moment of $ 1\rangle \leftrightarrow 2\rangle$	$2qa_0$ [12] ($q = 1.6 \times 10^{-19}$ C, $a_0 = 5.292 \times 10^{-11}$ m [20])
γ_{21}	Decay rate	6.066 MHz
γ_{31}, γ_{41}	Decay rates	$10^{-4} \cdot \gamma_{21}$ [18]
Ω_c	Rabi frequency (coupling laser)	40 MHz [18, Fig. 4]
μ_{RF}	Dipole moment of RF transition	$0.4899qa_0Q_n$ [18, Eq. (11)] $Q_n = 10000$ [18, Fig. 5 & Fig. 7]
Δ_{RF}	Detuning (signal/LO locked)	0 (on-resonant) [43]
N_{part}	Number of particles	300
N_{iter}	Number of iterations	50

- The other parameters are: $A = A_0 = 10$ mV/m; $M = 1$; $\Delta_{c,1} = 10$ kHz; $L = 2\lambda = 29.55$ cm; $\varphi = \varphi_0$.

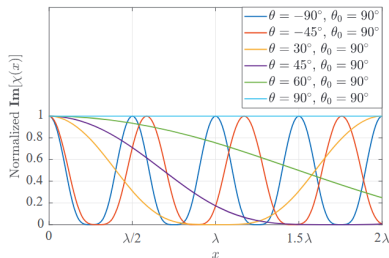


Fig. 4: Normalized $\text{Im}[\chi(x)]$ versus x , where x denotes the location in the atomic vapor as illustrated in Fig. 3. As can be observed, the susceptibility depends on x due to the interference between E_{SIG} and E_{LO} .

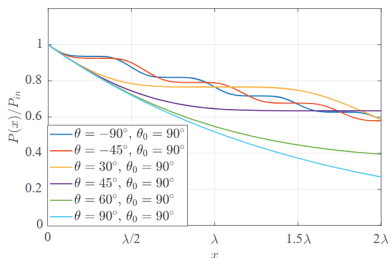


Fig. 5: The probe laser power at the length of x penetrating through the atomic vapor $P(x)$ versus x . As can be observed, $dP(x)/dx$ depends on x due to the interference between E_{SIG} and E_{LO} .

Determining the AoA of LO(1)

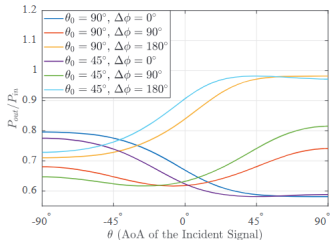


Fig. 6: P_{out}/P_{in} versus θ for different values of θ_0 and $\Delta\phi$.

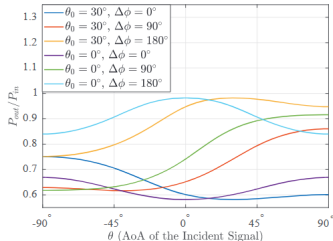


Fig. 7: P_{out}/P_{in} versus θ for different values of θ_0 and $\Delta\phi$.

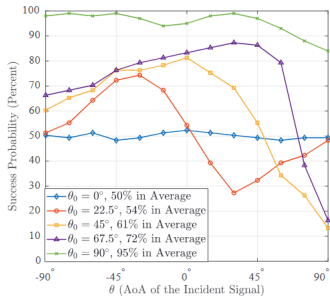


Fig. 8: Success probability of AoA detection versus θ under different θ_0 .

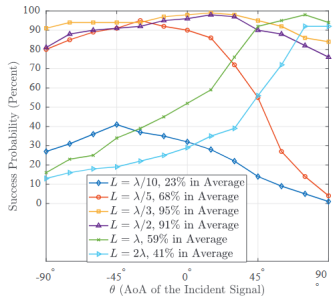


Fig. 9: Success probability of AoA detection versus θ for different L values.

- ▶ It is difficult to perform unambiguous AoA detection with only one measurement ($M = 1$).
- ▶ Evaluate the probability of successful AoA estimation versus the AoA θ for different values of the LO's AoA θ_0 .
- ▶ Note that the AoA estimation is considered to be successful if the corresponding root mean square error (RMSE), defined as

$$\text{RMSE} = \sqrt{\mathbb{E} \left\{ (\hat{\theta} - \theta)^2 \right\}}, \quad (28)$$

is less than 1° considering $N_{\text{iter}} = 50$ iterations for the proposed PSO-based estimation under noiseless conditions.

- ▶ Other parameters are set to: $M = 1$; $\Delta_{c,m} = 10$ kHz; $N_{\text{part}} = 300$; $w = 0.5$; $c_1 = 0.5$; $c_2 = 0.5$.

- Now we try to determine θ_0 to maximize the probability of successful AoA estimation.
- According to the following derivation, to achieve a high success probability, the following condition must be satisfied: $|2 \sin \theta_0 - \sin \theta| > 1$. In order to maximize the detectable range of θ , we need maximize the right-hand side upper bound above, i.e., $\arcsin(2 \sin \theta_0 - 1)$. Therefore, the optimal AoA of the LO signal is $\theta_0 = 90^\circ$.

$$\Omega_{RF}(x) = \frac{\sqrt{A_0^2 + A^2 + 2A_0A \cos [(k \sin \theta_0 - k \sin \theta)x - \Delta\phi]}}{h} \mu_{RF} \quad (29)$$

$$k \sin \theta_0 - k \sin \theta^{(\text{error})} = -(k \sin \theta_0 - k \sin \theta) \quad (30)$$

$$\sin \theta_0 - \sin \theta^{(\text{error})} = -\sin \theta_0 + \sin \theta \quad (31)$$

$$\sin \theta^{(\text{error})} = 2 \sin \theta_0 - \sin \theta \quad (32)$$

$$\theta^{(\text{error})} = \arcsin(2 \sin \theta_0 - \sin \theta) \quad (33)$$

Analysis

L determines the significance of interference effects and the non-convexity of the optimization. Too small ($L \approx \lambda/10$) gives weak interference; too large ($L \geq \lambda/2$) makes the transmittance $T(\theta)$ highly non-monotonic, increasing the risk of converging to wrong minima .

Recommended Selection

Recommended: $L \approx \lambda/3$. This balances interference strength and non-convexity.

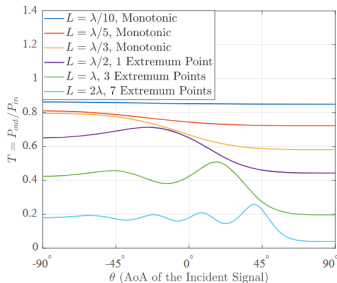


Fig. 10: Probe laser transmission $T = P_{out}/P_{in}$ versus the AoA of the incident signal θ for different lengths L of the atomic vapor cell.

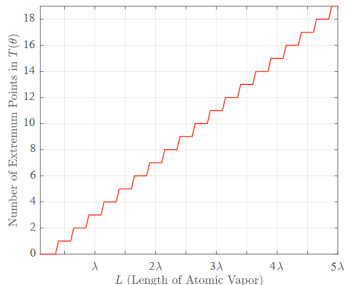


Fig. 11: The number of extremum points in $T(\theta)$ (as illustrated in Fig. 10) versus the length L of the atomic vapor cell.

Determining the Frequency Detuning of Coupling Laser

Analysis

Δ_c is a tunable parameter for obtaining multiple measurements. The variation of transmittance T with θ is most pronounced within a certain range of Δ_c , which benefits AoA estimation.

Recommended Range

Suggested: Δ_c between 1 kHz and 100 kHz.

At $\Delta_c = 10$ kHz, average success probability $\approx 95\%$.

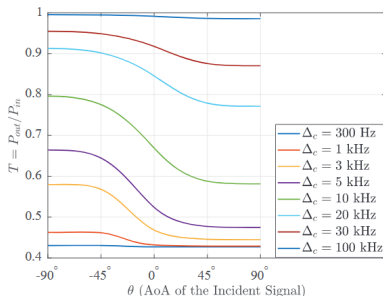


Fig. 12: Probe laser transmission $T = P_{out}/P_{in}$ versus the AoA of the incident signal θ for different coupling laser frequency detunings Δ_c .

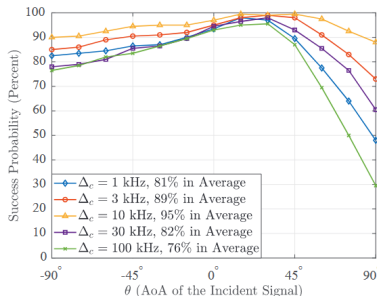


Fig. 13: The success probability of AoA detection versus θ for different coupling laser frequency detunings Δ_c .

Analysis

Increasing M (by scanning Δ_c) enables joint estimation of $\{A, \theta, \Delta\phi\}$, effectively resolving estimation ambiguities caused by non-injective mapping of $T(\theta)$ for $M = 1$ (see Figs. 6–7) and suppressing noise.

Effect

More measurements ($M > 1$) significantly improve success rate and accuracy. Even with $L = \lambda/3$, $M = 1$ already gives 95% success; For $M \geq 10$ and $\text{SNR} \geq 60$ dB, AoA RMSE $< 1^\circ$.

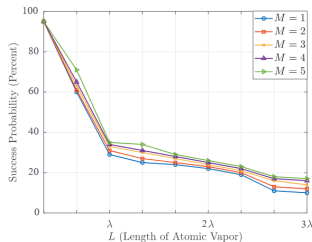


Fig. 14: The success probability of AoA detection versus L for different numbers of measurements M .

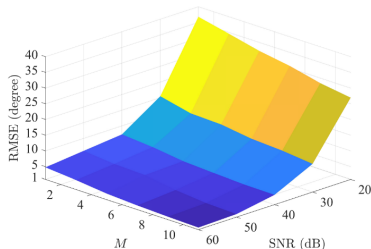


Fig. 15: The RMSE of θ versus the SNR and the number of measurements M .

Analysis

The residual function $R(A, \theta, \Delta\phi)$ is highly non-convex, requiring sufficient particles and iterations for the PSO algorithm to converge near the global optimum .

Recommended Settings

Use a large number of particles ($N_{\text{part}} \geq 200$) and sufficient iterations ($N_{\text{iter}} \approx 30$).

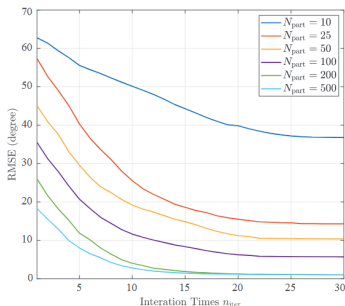


Fig. 18: The RMSE versus the iteration times n_{iter} under different numbers of particles.

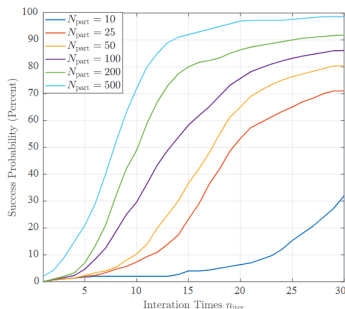


Fig. 19: The success probability versus the iteration times n_{iter} under different numbers of particles.

Scheme	Number of Receivers / Tags	Algorithm / Basis
Single Rydberg (Proposed)	1 Rydberg atom	Internal vapor interference + PSO
Twin Rydberg [10]	2 Rydberg atoms	Experimental array [10]
Backscattering [35]	1 conventional receiver + 5 tags	Backscatter-aided [35]
1/2/4/8 Conventional	1/2/4/8 conventional receivers	MUSIC algorithm [44]

Note: “1 Conventional” cannot perform AoA estimation and is treated as random guess .

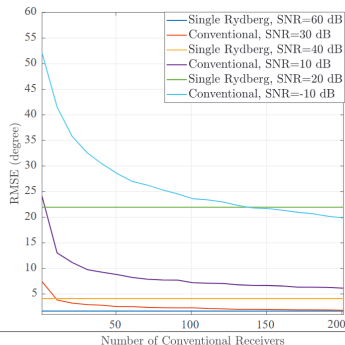
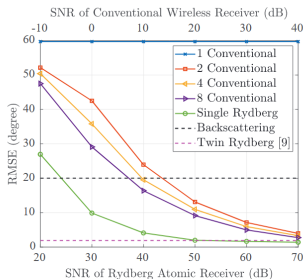


Table 1: Comparison of AoA Estimation Accuracy [1]

Scheme	N	RMSE (relative performance)
Single Conventional	1	Random guess (no estimation)
2 Conventional (MUSIC)	2	Higher than Single Rydberg
4 Conventional (MUSIC)	4	Higher than Single Rydberg
8 Conventional (MUSIC)	8	Higher than Single Rydberg
Twin Rydberg [10]	2	Slightly higher than Single Rydberg (at high SNR)
Single Rydberg (Proposed)	1	Lowest RMSE

The Single Rydberg scheme achieves better accuracy with only one receiver, demonstrating both hardware efficiency and superior performance .



SNR Advantage of Rydberg Receivers

To ensure a fair comparison, simulations assume that the Rydberg receiver achieves a **30 dB higher SNR** than conventional receivers .

Theoretical Justification (Appendix C)

- ▶ **Conventional receiver:** Dominated by **thermal noise** from electronic random motion, leading to high noise equivalent power (NEP).
- ▶ **Rydberg atom receiver:** Based on energy-level transitions induced by incident EM waves; **inherently immune to thermal noise** . The main noise source is **photon-shot noise** from the photodetector.
- ▶ With probe laser power $100\ \mu\text{W}$ and electric field $20\ \text{mV/m}$, the Rydberg receiver SNR can be $\approx 30.9\ \text{dB}$ higher than conventional receivers .

Received Power:

$$P_{\text{conventional}} = \frac{\lambda_{\text{RF}}^2}{4\pi} \cdot \frac{|E_{\text{RF}}|^2}{2Z_0}, \quad (34)$$

where λ_{RF} is the carrier wavelength, $|E_{\text{RF}}|$ the RF electric field amplitude, and $Z_0 = 120\pi \Omega$ the free-space impedance.

Thermal Noise Power:

$$P_{\text{thermal}} = k_B T_{\text{room}} \Delta f, \quad (35)$$

with $k_B = 1.38 \times 10^{-23} \text{ J/K}$, $T_{\text{room}} = 290 \text{ K}$, and Δf the signal bandwidth.

Total SNR (with LNA):

$$\text{SNR}_{\text{conventional}} = \frac{P_{\text{conventional}}}{\text{NF}_{\text{LNA}} P_{\text{thermal}}} = \frac{\lambda_{\text{RF}}^2 |E_{\text{RF}}|^2}{8\pi Z_0 k_B T_{\text{room}} \Delta f \text{NF}_{\text{LNA}}}. \quad (36)$$



Dominant Noise Source: Recent experiments [14, 17] show that the noise floor is dominated by **photon-shot noise** from the photodetector; other noise sources are negligible.

SNR Expression:

$$\text{SNR}_{\text{Rydberg}} = \frac{I_p}{2q\Delta f}, \quad (37)$$

where I_p is the photocurrent and $q = 1.6 \times 10^{-19}$ C.

Photocurrent Calculation:

$$I_p = \frac{P_{\text{pr}}\eta q\lambda_{\text{pr}}}{hc}, \quad (38)$$

with P_{pr} the probe laser power, η the quantum efficiency, λ_{pr} the probe wavelength, h Planck's constant, c the speed of light.

Final SNR:

$$\text{SNR}_{\text{Rydberg}} = \frac{P_{\text{pr}}\eta\lambda_{\text{pr}}}{2hc\Delta f}. \quad (39)$$

SNR Gain Formula:

$$\frac{\text{SNR}_{\text{Rydberg}}}{\text{SNR}_{\text{conventional}}} = \frac{4\pi P_{\text{pr}} \eta \lambda_{\text{pr}} Z_0 k_B T_{\text{room}} \text{NF}_{\text{LNA}}}{hc \lambda_{\text{RF}}^2 |E_{\text{RF}}|^2}. \quad (40)$$

Typical Parameters:

$$\begin{aligned} P_{\text{pr}} &= 100 \mu\text{W} [48], & \eta &= 65\% [14], \\ \lambda_{\text{pr}} &= 780 \text{ nm} [18], & \text{NF}_{\text{LNA}} &= 10^{0.35} \approx 2.24 \text{ (3.5 dB)} [51], \\ \lambda_{\text{RF}} &= c/f_c \approx 0.147 \text{ m } (f_c \approx 2.03 \text{ GHz}) [18], & |E_{\text{RF}}| &= 20 \text{ mV/m}. \end{aligned} \quad (41)$$

Result: Plugging these values into Eq. (40) yields a gain of approximately 30.9 **dB**.

1. Introduction
2. Rydberg Receiver Model
3. AOA Detection Model
4. Particle Swarm Optimization (PSO)-based Algorithm
5. Simulation Results
6. Future Work



- ▶ 推导叠加了线性磁场的磁化率表达式，引入多接收机建模 OFDM-MIMO。
- ▶ 推广外差场或里德堡检测通道进行多信号源 AoA 估计或结合其他量子增强感知任务。

Thank you!